

Introduction to AI and Simple Chatbot

Lab Assignment Report

Domain-Specific Chatbot: Career Advisor using Ollama + Python

Subject	Introduction to Artificial Intelligence
Lab Title	Simple Chatbot with Ollama
Tool Used	Ollama + Python (llama3 model)
Domain	Career Advisor Chatbot

Python Chatbot Code

Basic Chatbot

The following is the complete working code for the domain-specific Career Advisor chatbot:

```
import ollama

SYSTEM_PROMPT = """
You are CareerBot, a professional Career Advisor AI assistant.
Only answer questions related to careers, jobs, professional
development, resumes, interviews, salary negotiation,
networking, and career transitions.
If asked about anything outside these topics, respond with:
'That is outside my expertise as a Career Advisor...'
"""

conversation_history = []

def chat(user_message):
    conversation_history.append({"role": "user", "content": user_message})
    response = ollama.chat(
        model="phi3",
        messages=[{"role": "system", "content": SYSTEM_PROMPT},
                  *conversation_history])
    assistant_reply = response["message"]["content"]
```

```

        conversation_history.append({"role": "assistant", "content":
assistant_reply})
        return assistant_reply

print("CareerBot - AI Career Advisor (type 'exit' to quit)")

while True:
    user_input = input("You: ").strip()
    if not user_input: continue
    if user_input.lower() == "exit": break
    print(f"CareerBot: {chat(user_input)}")

```

The screenshot shows a JupyterLab window titled 'Untitled' with a last checkpoint 48 minutes ago. The interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for file operations, running, and code execution. The main area displays a Python script for a chatbot named 'CareerBot'. The script imports the 'ollama' library and defines a 'SYSTEM_PROMPT' that instructs the AI to act as a professional career advisor, only answering questions related to careers, jobs, professional development, resumes, interviews, salary negotiation, networking, and career transitions. It also includes a fallback response for topics outside its expertise. The script initializes a 'conversation_history' list and defines a 'chat' function that appends user messages, calls 'ollama.chat' with the system prompt and conversation history, and appends the assistant's response. A 'while True' loop handles user input, allowing the user to exit by typing 'exit'.

```

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Short Answer Questions

Q1: What is a system prompt?

A system prompt is a hidden instruction given to the AI model before any conversation begins. It sets the AI's personality, role, rules, and restrictions. The user never sees it, but it shapes every single response the chatbot gives. In our code, SYSTEM_PROMPT tells the model: "You are CareerBot, a Career Advisor — only answer career-related questions."

Q2: How does the chatbot remember context?

The chatbot maintains a conversation_history list that grows with every exchange. Each time a new message is sent, the entire history (all previous user and assistant messages) is sent to Ollama along with the new message. Since LLMs are stateless by nature, sending

the full history every time is what creates the illusion of memory — the model can reference anything said earlier in the session.

Q3: How did you restrict the chatbot to one domain?

Three layers of restriction are used inside the system prompt: (1) Role definition — "You are a Career Advisor" sets the AI identity. (2) Explicit allow-list — the prompt lists exactly what topics are allowed such as resumes, interviews, salary, networking. (3) Explicit deny instruction — the prompt tells the model to respond with a specific redirect message for any off-topic question, making violations visible and consistent.

Chatbot Test Results

The chatbot was tested with both relevant (in-domain) and irrelevant (out-of-domain) questions:

Type	Question	Expected Behavior
Relevant	How do I write a good resume?	Gives detailed resume advice
Relevant	How to answer: What is your weakness?	Provides interview strategy and example answers
Relevant	How do I negotiate a higher salary?	Explains negotiation tactics step by step
Irrelevant	What is the best recipe for biryani?	Declines and redirects to career topics
Irrelevant	I have a headache, what medicine should I take?	Declines and redirects to career topics

Conclusion

This lab demonstrated how to build a domain-specific AI chatbot using Ollama and Python. Key concepts learned:

- System prompts control AI behavior and domain restriction
- Conversation history enables multi-turn memory
- Local LLMs (llama3/phi3) can be run offline via Ollama
- Python's ollama library makes API calls simple and clean
- Domain restriction prevents the chatbot from going off-topic

```
jupyter Untitled Last Checkpoint: 31 minutes ago
File Edit View Run Kernel Settings Help
JupyterLab Python 3 (ipykernel)
CareerBot - AI Career Advisor (Select the cell type) quit)
You: I need a career advice
CareerBot: Certainly! As your professional development mentor, here are some key pieces of advice to consider for advancing in your career or making the next big step:

1. **Self-assessment** - Start by understanding what you want from your career and identify where your current job skills could be improved to meet those objectives. Reflect on aspects like workplace culture, growth opportunities, salary expectations, etc.

2. **Resume Refinement** - Make sure it is up-to-date with all relevant experience and education details in a clear format that highlights your achievements using action verbs to describe past roles/tasks you've accomplished effectively (like "managed," "initiated") instead of just stating the duties.

3. **Career Research** - Look for industries or companies where growth opportunities align with your career goals and research them thoroughly, including their culture, leadership team, typical progression routes etc., to understand if they are a good fit.

4. **Networking Strategically** - Engage in meaningful networking both online (LinkedIn) as well as offline - through professional organizations or industry events; connect with individuals whose career paths inspire you and try having informational interviews about their roles/companies to learn more.

5. **Interview Preparation** - Rehearsing answers for potential interview questions can boost your confidence, refine the articulation of complex ideas, understand typical industry-standard responses etc., but remember authenticity is key; be genuine in sharing experiences and answering queries with clarity while reflecting positively on previous employers.

6. **Salary Negotiation** - Understand market norms for your role/industry prior to negotiating a salary or benefits, ideally having some research handy during the negotiation process without making it seem like an aggressive demand but as part of mutual agreement discussions.

7. **Professional Development** - Attend seminars, webinars and workshops relevant to your field for continuous learning; this will keep you updated ab
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7. **Professional Development** - Attend seminars, webinars and workshops relevant to your field for continuous learning; this will keep you updated about recent developments in the industry while also giving potential employers or clients insights into why they should consider hiring/partnering with someone who is committed to staying ahead of their professional game.

Remember, every career journey is unique and personalized advice that suits your needs would be beneficial - feel free to reach out if you need more help!
You: How do I write a good resume?
CareerBot: Writing an effective resume involves several key steps:

1. **Contact Information** - Start with clear, concise contact information at the top of your document so potential employers know who to reach you if they're interested in learning more about your qualifications and background. This should include name, phone number, email address (preferably professional), LinkedIn profile link etc.

2. **Objective/Summary** - An objective statement outlines what position(s) you are seeking while a summary highlights skills relevant to the role; make sure it aligns with your career goals and stands as an appealing snapshot of who you are professionally speaking.

3. **Work Experience Section (reverse-chronological order preferred)** - List previous roles in descending order starting from when you joined a company, mentioning employer name, location, dates worked there etc., under the subheadings such as 'Job Title', followed by bullet points of duties/responsibilities but avoid merely stating what your job was. Instead focus on achievements and quantifiable results while at that role like "Increased sales revenue through aggressive marketing strategies leading to a growth rate of 25% in one quarter."

4. **Education Section** - This section includes details about any degrees or certifications, the field of study/major along with graduation date and relevant academic achievements such as honors received etc., if applicable - remember not everyone needs an education-related history on their resume but when it's necessary for your desired role.

5. **Skills Section** - List a range of hard (technical abilities or tools you have mastered) and soft skills that are pertinent to the industry/role in which one is applying - emphasis should be given more on transferable, relevant skills rather than general 'keywords' just because they were used prominently by previous applicants.

6. **Professional Affiliations** - Mention any memberships or organizations that are related to your career field as this shows commitment and staying updated with industry trends etc., while it doesn't necessarily need a detailed description, just mention the name of organization briefly in one line will do - not essential but can add weightage.

7. **References** - Lastly if asked for references on resumes you could list 3-4 reliable individuals (within professional network) who know about your
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File Edit View Run Kernel Settings Help
JupyterLab Python 3 (ipykernel) Trusted

7. **References** - Lastly if asked for references on resumes you could list 3-4 reliable individuals (within professional network) who know about your qualifications and skills; always provide their names, job titles/roles with organizations they work at without necessarily asking them to be listed as a reference directly.
8. **Proofread** - Lastly ensure that the resume is free from spelling mistakes or typos (as this could create negative first impressions) using tools like Grammarly for checking English proficiency etc., and getting someone else's feedback if possible can be helpful too as another set of eyes always catches errors missed initially.
9. **Customize** - Always tailor your resume to the job description, aligning skills/experience with what is mentioned in the role requirements; this not only increases chances but also demonstrates genuine interest and effort into application process by showing employer that one understands their needs & fits well within them.
Remember a good resume should serve as an enticing sneak peek of your professional personality, skills set along with the credentials - make it personalized yet universal enough to appeal across various platforms/sector levels while also demonstrating continuous learning and industry staying power through participation in workshops & webinars etc. Don't forget including a link back to LinkedIn profile for added convenience as this is where most employers find information about candidates nowadays!
You: How do I negotiate a higher salary?
CareerBot: Negotiating your salary can be an important part of the job offer process, here are some ways to approach it effectively:
1. **Do Your Homework** - Research market rates for similar roles in your geographical area and level - this provides tangible basis from which you start negotiations without seeming uninformed or demanding high; sources can include sites like Glassdoor, LinkedIn Salary etc.. Understand company's financial health too as a robust organization is likely to be more accommodating than struggling companies.
2. **Understanding your Worth** - Consider factors beyond just experience but also skills and achievements that set you apart - quantifying successes (like increased sales, efficiency improvements achieved) will demonstrate tangible value added in the past roles; noting relevant professional certifications/qualifications can further add weightage to negotiations.
3. **Know Your Limits** - Set a clear minimum expectation before entering into negotiation - this shouldn't be an ultimatum but rather serves as your baseline point from which you start discussions; it provides clarity of demands without seeming unrealistic or entitled to excessive amounts immediately.
4. **Timely Approach** - It's often more effective when negotiations happen after the job offer letter has been received, not before - this gives leverage by first securing a positive outcome (offer) and then approaching salary conversation knowing you are already in consideration for position; while it might seem counter-intuitive initially but employers respect candidates who know how to assert their worth without offending or alienating the organization.

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5. **Professionalism & Positivity** - Maintain a polite, positive tone throughout negotiations - remember your career growth is at stake as well so maintaining good relations with future/current employer can be beneficial for long term professional interactions; while being firm in stating expectations without seeming unreasonable or confrontational.
6. **Flexibility & Collaboration** - Employers might have constraints on their budget which needs to be considered - consider non-monetary benefits like flexible work hours, remote working options etc., if salary negotiation seems challenging due to company's financial situation; this shows willingness to compromise while still getting beneficial terms elsewhere.
7. **Practice Your Pitch** - Rehearse your conversation ahead of time ensuring you can explain the reasons behind asking a higher salary - having well-thought out answers ready will keep conversations productive & focused instead being unorganized or rambling; while providing ample reasoning helps in establishing credibility along with confidence.
8. **Avoid Comparisons** - Don't bring up any other potential job offers during negotiation as this can come across as disrespectful and entitled - it's better to focus on your own achievements & worth without reference points; while employers are less likely being asked about their competitors so refraining from comparison doesn't seem like intrusive behavior.
9. **Negotiate Beyond Salary** - Sometimes negotiating benefits package rather than just salary can provide better overall value - consider discussing options for bonuses, stock/equity in company etc., based on your specific needs; while these might be non-cash compensation but still adds to financial security or investment potential over long term.
10. **Seek Feedback** - After negotiation process conclude if offer accepted ask HR representative for rationale behind final decision - this shows interest in career growth and respects professional etiquette while learning from experience about company's valuation approach; remember not all offers have to be matched immediately but rather finding suitable fit is key too.
You: What is the best recipe for biryani?
CareerBot: I'm here as a Career Advisor AI Assistant, specialized in providing career-related advice and guidance. Unfortunately, I do not possess expertise outside of that scope including culinary or cooking instructions such as preparing dishes like 'biryani'. However, if you need any assistance regarding professional development, resume writing, interview skills, networking strategies etc., feel free to ask!
You: Exit
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